

Task 3 Output

Color, k=5, svd

Image ID - weight pairs sorted in descending order of weights for each latent feature:				
Latent feature: 1				
(ID: 8582 , Weight: 0.025784573),	(ID: 8612 , Weight: 0.025666567),	(ID: 7298 , Weight: 0.025513362),	(ID: 7318 , Weight: 0.02544487),	(ID: 7290 , Weight: 0.02544487),
Latent feature: 2				
(ID: 1798 , Weight: 0.044912852),	(ID: 1802 , Weight: 0.0448418),	(ID: 1806 , Weight: 0.04418382),	(ID: 1282 , Weight: 0.043932207),	(ID: 1786 , Weight: 0.043932207),
Latent feature: 3				
(ID: 6452 , Weight: 0.059039086),	(ID: 6716 , Weight: 0.049613554),	(ID: 1260 , Weight: 0.048916128),	(ID: 1140 , Weight: 0.04845765),	(ID: 1692 , Weight: 0.04845765),
Latent feature: 4				
(ID: 6660 , Weight: 0.05067903),	(ID: 5572 , Weight: 0.048254784),	(ID: 6944 , Weight: 0.04615262),	(ID: 5408 , Weight: 0.04291123),	(ID: 5330 , Weight: 0.04291123),
Latent feature: 5				
(ID: 8570 , Weight: 0.066895165),	(ID: 2804 , Weight: 0.063075215),	(ID: 7888 , Weight: 0.060582165),	(ID: 7908 , Weight: 0.054198936),	(ID: 6956 , Weight: 0.054198936),
+ Code + Markdown				

Color, k=5, nnmf

Image ID - weight pairs sorted in descending order of weights for each latent feature:				
Latent feature: 1				
(ID: 1652 , Weight: 3.6223851038144113),	(ID: 1798 , Weight: 3.6206088406526904),	(ID: 1802 , Weight: 3.6155690709494817),	(ID: 1806 , Weight: 3.6006044109494817),	(ID: 1808 , Weight: 3.6006044109494817),
Latent feature: 2				
(ID: 6904 , Weight: 3.1379675229210013),	(ID: 4434 , Weight: 3.119852866535056),	(ID: 7794 , Weight: 2.9523504824532436),	(ID: 7734 , Weight: 2.90904109494817),	(ID: 7736 , Weight: 2.90904109494817),
Latent feature: 3				
(ID: 5904 , Weight: 2.950836715682272),	(ID: 5906 , Weight: 2.814683257949999),	(ID: 5918 , Weight: 2.7602172301479913),	(ID: 8582 , Weight: 2.7406301479913),	(ID: 8584 , Weight: 2.7406301479913),
Latent feature: 4				
(ID: 8894 , Weight: 2.5261165330416313),	(ID: 6860 , Weight: 2.5181483433267333),	(ID: 8120 , Weight: 2.4993865474437635),	(ID: 8086 , Weight: 2.474846109494817),	(ID: 8088 , Weight: 2.474846109494817),
Latent feature: 5				
(ID: 6442 , Weight: 2.486108485742311),	(ID: 3820 , Weight: 2.4679184583056752),	(ID: 6464 , Weight: 2.45274678921214),	(ID: 8662 , Weight: 2.427106109494817),	(ID: 8664 , Weight: 2.427106109494817),
+ Code + Markdown				

Color, k=5, lda

Image ID - weight pairs sorted in descending order of weights for each latent feature:				
Latent feature: 1				
(ID: 2504 , Weight: 0.99774593),	(ID: 2502 , Weight: 0.9977341),	(ID: 2610 , Weight: 0.997702),	(ID: 2612 , Weight: 0.9976423),	(ID: 2218 , Weight: 0.9976423),
Latent feature: 2				
(ID: 1496 , Weight: 0.9982115),	(ID: 1500 , Weight: 0.9982114),	(ID: 1616 , Weight: 0.9979046),	(ID: 1498 , Weight: 0.9974125),	(ID: 1612 , Weight: 0.9974125),
Latent feature: 3				
(ID: 8646 , Weight: 0.9977248),	(ID: 6648 , Weight: 0.9977114),	(ID: 3236 , Weight: 0.99763006),	(ID: 4816 , Weight: 0.9976277),	(ID: 1910 , Weight: 0.9976277),
Latent feature: 4				
(ID: 1632 , Weight: 0.99832803),	(ID: 1652 , Weight: 0.9983019),	(ID: 1470 , Weight: 0.9982862),	(ID: 1736 , Weight: 0.99828064),	(ID: 1852 , Weight: 0.99828064),
Latent feature: 5				
(ID: 7734 , Weight: 0.94716734),	(ID: 6904 , Weight: 0.90896255),	(ID: 2964 , Weight: 0.9043759),	(ID: 1744 , Weight: 0.899169),	(ID: 3064 , Weight: 0.899169),
+ Code + Markdown				

Color, k=5, kmeans

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Image ID - weight pairs sorted in descending order of weights for each latent feature:

Latent feature: 1
(ID: 7548 , Weight: 60.254295349121094),      (ID: 6440 , Weight: 59.987545013427734),      (ID: 5122 , Weight: 56.403255462646484),      (ID: 7466 , Weight: 55.33148:

Latent feature: 2
(ID: 7466 , Weight: 69.42273712158203),      (ID: 7548 , Weight: 63.98664474487305),      (ID: 7896 , Weight: 63.279544830322266),      (ID: 6452 , Weight: 61.88714:

Latent feature: 3
(ID: 7466 , Weight: 68.66119384765625),      (ID: 7548 , Weight: 67.84347534179688),      (ID: 6440 , Weight: 66.5665054321289),      (ID: 5122 , Weight: 66.56352:

Latent feature: 4
(ID: 6440 , Weight: 68.2012939453125),      (ID: 7548 , Weight: 67.1377182006836),      (ID: 5122 , Weight: 66.38748168945312),      (ID: 3484 , Weight: 63.55163:

Latent feature: 5
(ID: 7548 , Weight: 65.12383270263672),      (ID: 6440 , Weight: 63.821922302246094),      (ID: 5122 , Weight: 60.4796142578125),      (ID: 7794 , Weight: 59.468444
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resnet, k=5, svd

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Image ID - weight pairs sorted in descending order of weights for each latent feature:

Latent feature: 1
(ID: 800 , Weight: 0.0531554004),      (ID: 812 , Weight: 0.052846022),      (ID: 32 , Weight: 0.052470617),      (ID: 230 , Weight: 0.052324735),      (ID: 16 , Wei

Latent feature: 2
(ID: 748 , Weight: 0.032626193),      (ID: 742 , Weight: 0.03047488),      (ID: 740 , Weight: 0.030232871),      (ID: 618 , Weight: 0.030127546),      (ID: 436 , We

Latent feature: 3
(ID: 3528 , Weight: 0.05452447),      (ID: 3638 , Weight: 0.05292779),      (ID: 3618 , Weight: 0.052419327),      (ID: 3634 , Weight: 0.050650384),      (ID: 3606 , W

Latent feature: 4
(ID: 6574 , Weight: 0.061276175),      (ID: 6580 , Weight: 0.060785174),      (ID: 3590 , Weight: 0.060263947),      (ID: 3894 , Weight: 0.057363026),      (ID: 6588 , W

Latent feature: 5
(ID: 2292 , Weight: 0.058403146),      (ID: 2518 , Weight: 0.057290234),      (ID: 5556 , Weight: 0.055316433),      (ID: 5608 , Weight: 0.054794457),      (ID: 2602 , W
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resnet, k=5, nnmf

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Image ID - weight pairs sorted in descending order of weights for each latent feature:

Latent feature: 1
(ID: 5772 , Weight: 3.44873160560374),      (ID: 5756 , Weight: 3.4274944853556573),      (ID: 5740 , Weight: 3.26444717455070857),      (ID: 3042 , Weight: 3.247373:

Latent feature: 2
(ID: 3466 , Weight: 2.8974072795155443),      (ID: 8628 , Weight: 2.8509915206725234),      (ID: 3192 , Weight: 2.766543488978331),      (ID: 7498 , Weight: 2.764734:

Latent feature: 3
(ID: 5616 , Weight: 2.672731081787181),      (ID: 3234 , Weight: 2.6289028886963197),      (ID: 2952 , Weight: 2.551903506134945),      (ID: 7724 , Weight: 2.514444:

Latent feature: 4
(ID: 1120 , Weight: 2.9892287240717943),      (ID: 1798 , Weight: 2.97150216444606265),      (ID: 1634 , Weight: 2.9671359876311185),      (ID: 1642 , Weight: 2.942288:

Latent feature: 5
(ID: 514 , Weight: 3.625806407934092),      (ID: 808 , Weight: 3.62408086723564565),      (ID: 742 , Weight: 3.586018924077069),      (ID: 618 , Weight: 3.57418231
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resnet, k=5, lda

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Image ID - weight pairs sorted in descending order of weights for each latent feature:

Latent feature: 1
(ID: 6454 , Weight: 0.99646634),      (ID: 6572 , Weight: 0.9964548),      (ID: 7478 , Weight: 0.9964449),      (ID: 6402 , Weight: 0.9964481),      (ID: 4528 , W

Latent feature: 2
(ID: 748 , Weight: 0.99677354),      (ID: 618 , Weight: 0.99672425),      (ID: 514 , Weight: 0.99670243),      (ID: 614 , Weight: 0.99669796),      (ID: 616 , We

Latent feature: 3
(ID: 1786 , Weight: 0.93033594),      (ID: 1446 , Weight: 0.92098594),      (ID: 1642 , Weight: 0.91226214),      (ID: 1354 , Weight: 0.9101634),      (ID: 1686 , W

Latent feature: 4
(ID: 4258 , Weight: 0.8996958),      (ID: 4148 , Weight: 0.8322446),      (ID: 4252 , Weight: 0.81050766),      (ID: 2908 , Weight: 0.7981833),      (ID: 7414 , W

Latent feature: 5
(ID: 5812 , Weight: 0.8094199),      (ID: 5770 , Weight: 0.8064508),      (ID: 5726 , Weight: 0.7966189),      (ID: 3952 , Weight: 0.7958496),      (ID: 5750 , W
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resnet, k=5, kmeans

Image ID - weight pairs sorted in descending order of weights for each latent feature:			
Latent feature: 1			
(ID: 748 , Weight: 72.44503021240234),	(ID: 740 , Weight: 68.55811309814453),	(ID: 2952 , Weight: 67.23258972167969),	(ID: 3236 , Weight: 67.11959011000000),
Latent feature: 2			
(ID: 748 , Weight: 68.23404693603516),	(ID: 3236 , Weight: 63.89088821411133),	(ID: 2952 , Weight: 63.84645080566406),	(ID: 5812 , Weight: 63.02061000000000),
Latent feature: 3			
(ID: 2952 , Weight: 75.11670684814453),	(ID: 5772 , Weight: 73.74195861816406),	(ID: 2920 , Weight: 73.1340560913086),	(ID: 5736 , Weight: 72.51091000000000),
Latent feature: 4			
(ID: 748 , Weight: 72.80250549316406),	(ID: 2952 , Weight: 70.6156997680664),	(ID: 740 , Weight: 70.01309204101562),	(ID: 5812 , Weight: 69.05281000000000),
Latent feature: 5			
(ID: 748 , Weight: 70.24055480957031),	(ID: 5812 , Weight: 67.50411987304680),	(ID: 2952 , Weight: 66.6607666015625),	(ID: 3236 , Weight: 65.11505000000000),